



DAYANANDA SAGAR
UNIVERSITY



Research Milestone Roadmap

Department of Computer Science & Medical Engineering

School of Engineering, Dayananda Sagar University

June–August 2026 Action Plan

A professional operating blueprint for high-quality research, IEEE/Springer/Scopus submissions, Q1/Q2 journal pipeline, open research infrastructure, and department-level digital branding.



Official references
included in Section 2

Working motto: *Innovating Intelligence for Human Health.*

Strategic identity: CSME Research Cell for AI-Driven Medical Engineering and Digital Health.

Document principle: one roadmap, one tracker, one evidence folder, one decision log - reducing redundancy in information sharing.

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1 Executive Milestone Snapshot

Purpose of this roadmap

This roadmap converts the department's first 3 months of research leadership into a disciplined milestone system. It is designed to make CS&ME visible as the AI-for-Healthcare research face of DSU-SoE while avoiding redundancy in communication, duplicate files, informal promises, and untracked publication work.



Q1/Q2 journal submissions by 31 Aug 2026



IEEE/Springer/Scopus manuscripts submitted or ready



Reproducible GitHub repositories with model cards



Professional branding posts across LinkedIn/X



Student research teams mapped to mentors



Internal paper review clinics



Project demo videos and research teasers



Clinical/industry collaboration meetings

Realistic expectation

By August 2026, the measurable success is not guaranteed acceptance in Q1/Q2 journals, because journal review cycles are outside departmental control. The measurable success is: submission-ready manuscripts, credible experiments, reproducible code, public-facing research branding, collaboration records, and a visible research pipeline.

2 Institutional Alignment

CS&ME mandate

The official CS&ME programme integrates Engineering and Medical Sciences, merging engineering, healthcare, and technology into an interdisciplinary programme. The page also emphasizes medical imaging, biomedical signal processing, Digital Healthcare, Medical AI, smart medical devices, telemedicine, real-world applications, and rural/public health impact.^[1]

DSU AI-first direction

DSU's AI-first initiative places AI at the core of education, research, innovation, entrepreneurship, and industry partnerships. Its listed research domains include Computer Vision and Healthcare AI, with healthcare/life sciences focus areas such as medical imaging, drug discovery, and diagnostics.^[2]

Infrastructure opportunity

DSU's NVIDIA AI Factory page highlights AI infrastructure for healthcare and biomedical AI solutions, including DGX B200 systems, Jetson edge devices, GPU workstations, PyTorch, TensorFlow, RAPIDS, and TensorRT.^[3]

Source notes

1. [DSU CS&ME official page](#): programme overview, collaborators, objectives and real-world impact emphasis.
2. [DSU AI-First page](#): AI at the core of learning, research and innovation; research domains including Healthcare AI.
3. [DSU NVIDIA AI Factory page](#): AI stack, DGX B200, Jetson, frameworks and healthcare/biomedical AI applications.

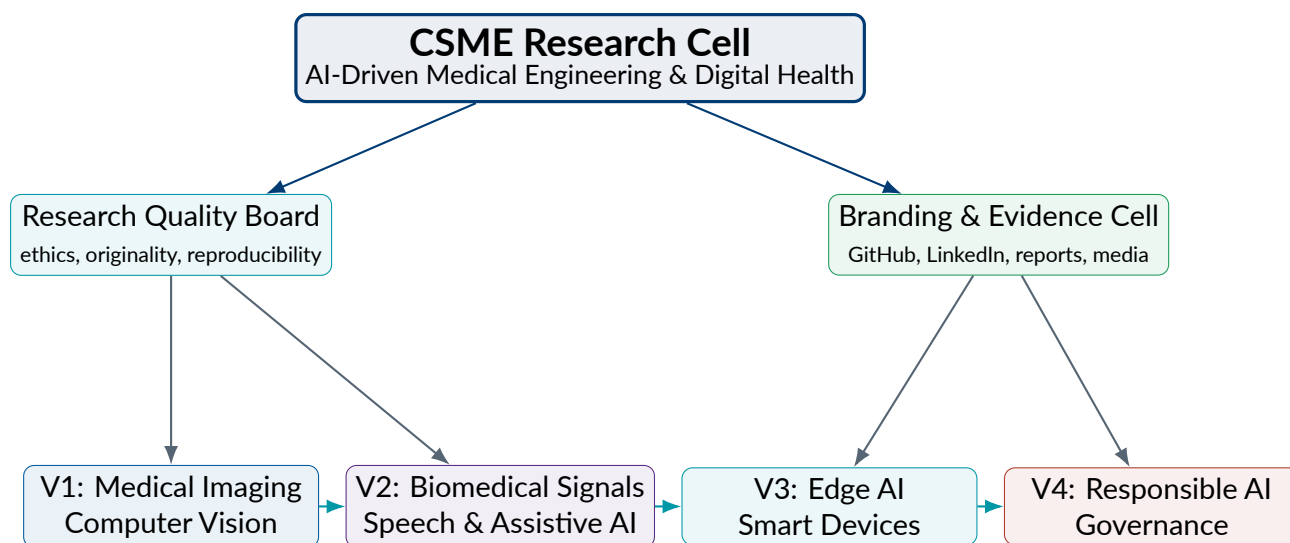
3 Research Identity and Operating Model

Proposed research identity

CSME Research Cell for AI-Driven Medical Engineering and Digital Health

Positioning statement: The Department of CS&ME will build ethically grounded, clinically meaningful, reproducible, AI-enabled engineering solutions for healthcare, with strong emphasis on medical imaging, biomedical signals, digital health, smart medical devices, assistive technologies, and translational impact.

3.1 Operating Architecture



3.2 Research Verticals

Vertical	Core focus	Immediate paper themes
V1. Medical Imaging, Computer Vision and Multimodal Diagnosis	X-ray, PET/CT, MRI, ultrasound, retinal imaging, fetal/infant imaging, radiomics, explainability.	PET/CT tumour analysis; newborn footprint-palmpoint quality; low-resource medical vision; explainable ViTs for diagnostics.
V2. Biomedical Signals, EEG, Speech and Assistive AI	EEG epilepsy analysis, speech impairment, cerebral palsy assistive communication, wearable physiological signals.	Multi-task EEG analysis; adaptive AAC; biomedical signal transformers; multimodal neuro-assistive AI.

V3. Edge AI, Smart Devices and AIoT Healthcare

Jetson/TinyML, medical IoT, smart devices, telemedicine, real-time health monitoring, infant biometrics.

Edge-AI infant identity; privacy-preserving health monitoring; TinyML medical capture quality; AIoT public health devices.

V4. Responsible AI, Medical Data Governance and Reproducible Research

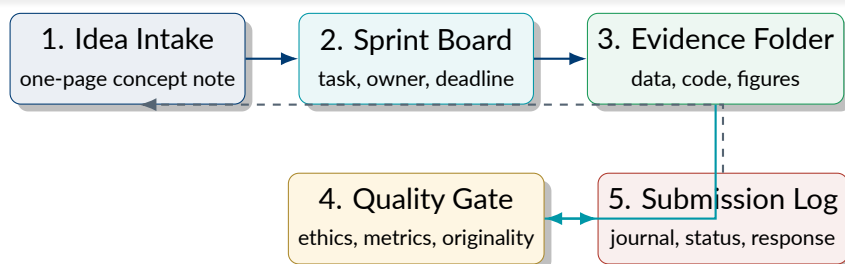
Ethics, privacy, bias, explainability, data leakage audit, reproducible pipelines, research integrity.

Responsible medical AI; leakage audit in medical imaging; model cards; clinical validation checklists.

4 Milestone Logic: How Redundancy Will Be Removed

Single-source operating rule

Every research activity must exist in exactly one official tracker. Every file must exist in exactly one evidence folder. Every decision must be recorded in exactly one decision log. Every publication must pass exactly one quality-gate checklist before submission.



4.1 Digital System of Record

Need	Recommended system	Rule
Research tracker	Google Sheet / Notion / Airtable named CSME Research Tracker 2026	Only this tracker determines official status.
Evidence storage	Google Drive / OneDrive: CSME Research Evidence	One folder per paper/project; no personal copies as official versions.
Code and reproducibility	GitHub Organization: DSU-CSME-Research	One repository per project; all figures and scripts versioned.
Meetings and decisions	Research Decision Log	Decisions are valid only when recorded with date, owner and action.
Branding	LinkedIn, X/Twitter and research newsletter calendar	Only approved posts go public; avoid fragmented personal announcements.

5 End-of-August Outcome Dashboard

Outcome Area	Milestone by 31 August 2026	Evidence to be archived
Research governance	CSME Research Cell constituted with verticals, review board, branding cell and weekly research clinic.	Charter PDF, membership list, meeting calendar, minutes.
High-quality journals	At least 2 Q1/Q2 journal manuscripts submitted or in final submission checklist.	Submission receipts, final manuscript PDFs, reviewer checklist.
IEEE/Springer/Scopus pipeline	At least 3 additional manuscripts targeted to IEEE/Springer/Scopus indexed routes.	Venue mapping sheet, manuscript drafts, author contribution logs.
Conference pipeline	At least 3 IEEE/Springer conference papers submitted or camera-ready for submission.	CFP archive, templates, submission confirmations.
Reproducible research	At least 3 repositories with README, license, environment file, notebooks, model card and result tables.	GitHub links, release tags, repository audit checklist.
Department branding	GitHub organization, LinkedIn page, X/Twitter handle, visual templates and monthly research digest.	Page URLs, screenshots, analytics, content calendar.
Student research involvement	5–6 student teams with faculty mentors and publication/demonstration deliverables.	Team list, mentor mapping, project cards, GitHub commits.
Visibility and outreach	12–15 branded posts, 3 demo videos, 1 brochure and 1 CSME AI-for-Healthcare Research Day.	Post links, brochure PDF, photographs, attendance report.

6 Three-Month Phase Plan

Phase principle

The department should not wait for perfect infrastructure before starting publications. Governance, branding, code infrastructure and manuscripts should run in parallel, but with a single tracker to prevent duplication.

Phase 1: Foundation and Research Audit

10–21 June 2026

Constitute the Research Cell, define verticals, audit faculty expertise, identify datasets, shortlist manuscripts and set up the single research tracker.

Phase 2: Branding Infrastructure and Manuscript Sprint 1

22 June–6 July 2026

Create GitHub organization, LinkedIn/X pages, repository standards, first branded posts and manuscript skeletons for P1–P4.

Phase 3: Experiments, Drafting and Public Visibility

7–21 July 2026

Run experiments, prepare diagrams, produce 60% drafts for top two papers, and launch research visibility campaign.

Phase 4: Manuscript Sprint 2 and Collaboration Activation

22 July–5 August 2026

Complete P1/P2 drafts, begin P3–P5 drafting, contact collaborators and release GitHub v1.0 repositories.

Phase 5: Submission and Branding Consolidation

6–20 August 2026

Submit P1/P2, submit P3/P4 to conferences where feasible, release brochure and publish August research newsletter.

Phase 6: Showcase, Review and September Pipeline

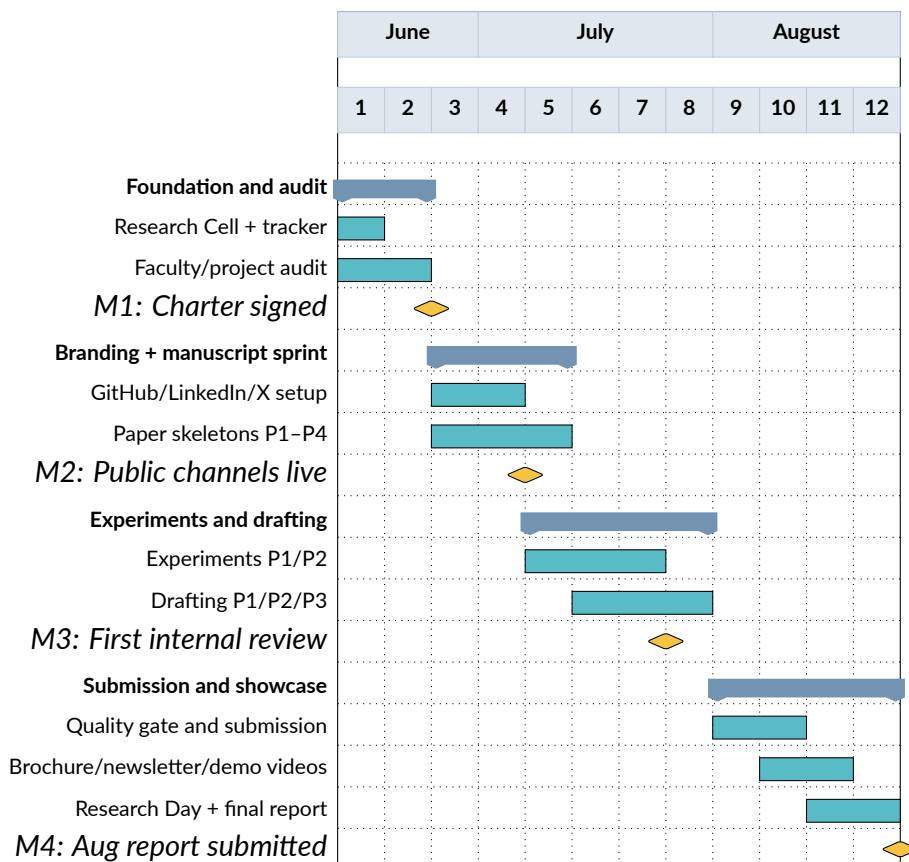
21–31 August 2026

Organize CSME AI-for-Healthcare Research Day, publish demo videos, finalize September pipeline and submit report to HoD/Dean.

7 Week-by-Week Milestone Calendar

Week	Core milestone	Research deliverable	Branding/communication deliverable	Review evidence
10–14 Jun	Research Cell launch and role clarity	Research Cell charter; vertical leads; author teams for first 5 papers.	Internal announcement; official folder architecture.	Signed charter; tracker created.
15–21 Jun	Research audit and project triage	Faculty expertise matrix; dataset inventory; paper pipeline P1–P8.	Standard naming rules; file-sharing SOP.	Audit sheet; gap list.
22–28 Jun	Branding infrastructure setup	Manuscript skeletons for P1–P4; venue shortlist.	GitHub org, LinkedIn page, X/Twitter handle; first three official posts.	Screenshots; repository shell.
29 Jun–5 Jul	Manuscript Sprint 1	Abstracts, contributions, method diagrams, experiment plan.	Banner templates; post template; newsletter template.	Internal review notes.
6–12 Jul	Experiment Sprint 1	Baselines for P1/P2; data split protocol; metrics plan.	Research spotlight post: Medical Imaging AI.	Result sheets; code commits.
13–19 Jul	Writing Sprint 1	60% draft for P1/P2; related work matrix.	Spotlight post: Infant Biometrics / Edge AI.	Drafts in evidence folder.
20–26 Jul	Technical review and revision	Ablation tables, statistical validation, limitations section.	Spotlight post: EEG/Assistive AI.	Reviewer checklist.
27 Jul–2 Aug	Collaboration activation	Clinical/industry meeting notes; P3–P5 50% drafts.	Partner/outreach post; GitHub release v0.1.	MoM and action list.
3–9 Aug	Submission preparation	P1 final submission package; P2 final package.	Pre-submission announcement; video script.	Quality gate sign-off.
10–16 Aug	First submission window	P1/P2 submitted or in final portal stage; P3/P4 conference submission.	Independence Day research post: AI for Bharat Healthcare.	Submission receipts.
17–23 Aug	Branding consolidation	P5/P6 draft completion; repository v1.0 release.	CSME Research Brochure; Newsletter Issue 1.	Brochure PDF; analytics.
24–31 Aug	Research showcase and reporting	CSME AI-for-Healthcare Research Day; September–December plan.	3 demo videos; report to HoD/Dean.	Event report; KPI dashboard.

8 Visual Timeline



9 Publication Milestone Portfolio

9.1 Priority Papers

ID	Manuscript theme	Venue class	Milestone by 31 Aug 2026	Evidence
P1	CAMELOT: Camouflage-Aware Motion-Stabilized PET/CT Tumour Segmentation and Stage-Oriented Reporting	Q1/Q2 journal	Full manuscript submitted or final portal-stage package completed.	Manuscript, figures, tables, code, response-free submission package.
P2	Edge-AI Newborn Footprint–Palmpoint Capture and Quality Assessment for Infant Identity Assurance	Q1/Q2 or IEEE Access/Springer journal	Full manuscript submitted/preprint ready; demo repository released.	Paper, device workflow, dataset protocol, GitHub v1.0.
P3	AI-Assisted Multi-Task EEG Analysis for Epilepsy and Related Neurological Conditions	IEEE/Springer conference or Q2 journal	Conference-ready paper with experiments and clinical interpretation plan.	Manuscript, model notebook, evaluation table.

P4	Adaptive Multimodal AAC for Children with Cerebral Palsy: Human-Centred AI Framework	Springer/IEEE conference	Submission completed or final technical draft.	Prototype notes, architecture, usability plan.
P5	Responsible AI Framework for Medical Engineering Education and Research	Scopus/Q2/Q3 journal or conference	Draft completed; literature matrix complete.	Literature sheet, draft, figure set.
P6	Data Leakage and Reproducibility in Medical Image Classification: Practical Audit Framework	Q2/Q3 journal/conference	Draft + reproducible audit repository ready.	Audit checklist, example repo, validation examples.
P7	AIoT-Based Edge Health Monitoring for Public and Rural Healthcare	IEEE/Springer conference	Abstract, architecture and experiment design completed.	Design brief, hardware plan, poster.
P8	AI-First Medical Engineering Education: Translational Framework for CS–Healthcare Integration	Scopus-indexed journal/conference	Literature review matrix and outline completed.	Bibliography, concept note, framework diagram.

9.2 Paper Sprint Template

Every paper must use this internal sprint structure

- Problem statement:** one paragraph, one clinical/engineering gap, one measurable objective.
- Contribution statement:** 3–5 contributions, no exaggerated claims.
- Dataset statement:** source, size, inclusion/exclusion criteria, split protocol and ethics/public-data note.
- Baselines:** at least 3 relevant baselines, including a simple classical baseline where appropriate.
- Ablation:** verify each methodological component contributes meaningfully.
- Statistics:** confidence intervals or significance testing where feasible.
- Reproducibility:** GitHub/controlled repository, environment file, README, model card.
- Limitations:** clinical, dataset, generalization and deployment limitations clearly stated.

10 Detailed Milestones

M0. Research Cell Charter Approved

Evidence: PDF charter, vertical leads, members, responsibility map and weekly clinic slot.

Owner/Review: Department Research Lead; HoD/Chairperson review.

M1. Single Research Tracker Operational

Evidence: Live tracker with paper ID, title, team, target venue, status, evidence folder, next action and deadline.

Owner/Review: Research Coordinator; all vertical leads update weekly.

M2. Digital Branding Channels Launched

Evidence: GitHub organization, LinkedIn page, X/Twitter handle, official bios, first 3 posts and banner assets.

Owner/Review: Branding Cell; approval by Research Lead/HoD.

M3. Publication Portfolio Frozen

Evidence: P1–P8 finalized with author teams, target venues and evidence folders.

Owner/Review: Quality Board; recorded in decision log.

M4. Experiment Protocols Locked

Evidence: Datasets, train-test splits, metrics, ablation protocol and ethical/data governance notes finalized for P1–P4.

Owner/Review: Vertical Leads; QA Board signs off.

M5. First Draft Review Completed

Evidence: P1/P2 60% drafts with literature review, methods, figures and preliminary results.

Owner/Review: Two internal reviewers per paper.

M6. GitHub v0.1 Released

Evidence: Minimum 2 repositories with README, project structure, environment, notebooks and citation file.

Owner/Review: Repository maintainer; reproducibility reviewer.

M7. Collaboration Meetings Completed

Evidence: At least 3 meetings with clinical/industry/internal partners and documented minutes.

Owner/Review: Research Lead; collaboration coordinator.

M8. Submission Quality Gate Passed

Evidence: P1/P2 pass originality, ethics, methodology, figure, reference and reproducibility checks.

Owner/Review: Quality Board; final sign-off.

M9. First Submission Window Completed

Evidence: P1/P2 submitted or in final journal portal stage; P3/P4 submitted or conference-ready.

Owner/Review: Corresponding authors; submission receipts archived.

M10. CSME Research Brochure and Newsletter Released

Evidence: Professional brochure, August newsletter and research social media analytics.

Owner/Review: Branding Cell; HoD/Dean copy.

M11. CSME AI-for-Healthcare Research Day Completed

Evidence: Posters, demos, invited talk, student presentations, photos and event report.

Owner/Review: Research Cell; student teams.

M12. Final August Report Submitted

Evidence: KPI dashboard, submissions, branding analytics, collaboration records and September–December plan.

Owner/Review: Research Lead; submitted to HoD/Dean.

11 Journal and Conference Strategy

Venue selection rule

Select the venue based on scope fit, paper maturity, APC availability, expected review cycle, indexing status, and ethical credibility. Do not choose journals only because the review is fast. Verify Scopus, Web of Science, SCImago quartile and publisher status immediately before submission.

Track	Use when	Suggested action
Q1/Q2 journal	Strong novelty, adequate experiments, clean dataset protocol, meaningful clinical/engineering contribution.	Use P1 and P2 for this route; prepare full response to scope, limitations and reproducibility.
IEEE Access / broad OA route	Interdisciplinary work, mature experiments, APC available, need for faster dissemination.	Consider only after quality gate; ensure novelty is clearly positioned.
Springer journal route	Interdisciplinary CS/AI/health informatics work with reproducible implementation.	Use P5/P6 or refined P3 where scope fits.
IEEE/Springer conference	Prototype work, early results, student engagement, rapid visibility.	Use P3/P4/P7; convert into journal extension later.
Review/tutorial paper	Strong literature matrix, departmental positioning, education/research framework.	Use P8 for department branding and thought leadership.

11.1 Publication Calendar

Period	Paper milestone	Action	Status colour
10–21 Jun	P1–P8 intake	Freeze topic, authorship, venue map and evidence folders.	Blue: setup
22 Jun–6 Jul	P1–P4 skeletons	Abstract, contributions, method diagrams, literature matrix.	Teal: drafting
7–21 Jul	P1/P2 experiments	Baselines, ablations, results, figures and limitations.	Green: evidence
22 Jul–5 Aug	P1/P2 full drafts	Internal review, revision, plagiarism/originality check.	Gold: quality gate

6–20 Aug	P1/P2 submission	Submit or complete portal-ready package; P3/P4 conference submission.	Red: deadline
21–31 Aug	P5/P6 next wave	Prepare September submissions; archive all outcomes.	Purple: pipeline

12 GitHub Organization and Reproducible Research Standard

Recommended organization name

DSU-CSME-Research

Bio: Official research GitHub organization of the Department of Computer Science & Medical Engineering, School of Engineering, Dayananda Sagar University. We build AI-driven, ethical, reproducible and translational healthcare technologies.

Pinned repositories

- medical-imaging-ai
- edge-ai-infant-biometrics
- biomedical-signals-ai
- assistive-ai-healthcare
- responsible-medical-ai
- csme-research-publications

Repository skeleton

- README.md, LICENSE
- CITATION.cff
- requirements.txt / environment.yml
- data/README.md
- notebooks/, src/, results/
- docs/, model_card.md

Release rule

No repository should be public if it contains patient-identifiable data, unpublished collaborator data, confidential clinical files, or unapproved institutional information. Use synthetic/demo data or controlled-access instructions when required.

13 Department Branding System

13.1 Channel Setup

Channel	Name/positioning	Content rule
LinkedIn Page	Department of CS & Medical Engineering, DSU-SoE	Professional visibility: achievements, research highlights, collaborations, student innovation, publications.
X/Twitter	@DSU_CSME or @DSU_AIHealth	Short updates, CFPs, accepted/-submitted papers, project demos, event threads.
GitHub	DSU-CSME-Research	Reproducible code, notebooks, project documentation and controlled research artefacts.
Newsletter	CSME Research Pulse	Monthly digest: papers, projects, students, collaborators, deadlines, GitHub repo of the month.

Brochure	CSME AI-for-Healthcare Research Brochure	6–8 page PDF for website, visitors, collaborators and student recruitment.
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13.2 Three-Month Content Calendar

Week	LinkedIn	X/Twitter	Evidence asset
W1	Introducing CSME Research Cell	Why AI for Healthcare at DSU?	Logo, mission statement.
W2	Four research verticals	Medical imaging + biomedical signals thread	Vertical infographic.
W3	GitHub organization launch	Reproducible research standard	Screenshot and repo links.
W4	Research spotlight: infant biometrics	Edge AI for clinical capture quality	30-sec concept video.
W5	Research spotlight: PET/CT AI	Q1/Q2 publication pipeline	Architecture figure.
W6	Research spotlight: EEG/Assistive AI	Clinical collaboration focus	Team photo / schematic.
W7	Student research teams	Student research culture	Mentor-team mapping.
W8	Responsible AI and reproducibility	Avoiding data leakage in medical AI	Checklist graphic.
W9	Submission sprint update	Publication quality gate	Quality gate card.
W10	AI for Bharat Healthcare	Rural/public health impact	Independence Day visual.
W11	CSME Research Pulse issue	Brochure release	Newsletter PDF.
W12	CSME AI-for-Healthcare Research Day	Demo video thread	Event report/video links.

14 Meeting Cadence and Communication Protocol

No-redundancy meeting design

Meetings should create decisions, not repeat information. Status updates must be written in the tracker before the meeting. Meeting time should be used only for blockers, decisions, quality review and next actions.

Cadence	Meeting	Agenda	Output
Weekly, 45 min	CSME Research Clinic	Paper progress, blockers, experiments, submissions, collaborator updates.	Updated tracker, action list, decision log.
Weekly, 30 min	Branding huddle	Upcoming posts, approvals, assets, analytics.	Content calendar updated.

Fortnightly, 60 min	Quality Board re-view	Manuscripts nearing submission, reproducibility, ethics, originality.	Quality gate sign-off or revision memo.
Monthly, 75 min	Research showcase	Student/faculty demonstrations, paper pitch, external invitee if possible.	Demo video, event notes, next pipeline.

14.1 Standard Meeting Note Template

Template

Date:

Participants:

Decisions taken:

1. Decision, owner, deadline.

Blockers:

Next actions:

Files/evidence updated:

Meeting:

15 RACI Matrix

Activity	Responsible	Accountable	Consulted	Informed
Research Cell charter	Research Lead	HoD/Chairperson	Vertical leads	Faculty/students
Publication tracker	Research Coordinator	Research Lead	All authors	HoD
P1/P2 journal submissions	Corresponding author	Research Lead	QA Board, co-authors	HoD/Dean
P3/P4 conference submissions	Paper team lead	Vertical lead	Student teams	Research Cell
GitHub organization	Repository maintainer	Branding Cell lead	QA/reproducibility reviewer	Department
LinkedIn/X posts	Branding Cell	Research Lead-/HoD	Paper teams	DSU community
Research brochure/newsletter	Branding Cell	HoD/Research Lead	Vertical leads	Dean/website team
Collaboration meetings	Collaboration coordinator	Research Lead	Clinical/industry partners	HoD/Dean
Research Day	Event coordinator	Research Lead-/HoD	All verticals	Students/faculty/partners

16 KPI Dashboard

KPI	June target	July target	August target	Evidence
Research Cell setup	Charter draft	Functional review board	Reported in Aug summary	Charter and minutes

Q1/Q2 journal papers	2 paper concepts	2 full drafts	2 submissions / portal-ready packages	Manuscripts, receipts
IEEE/Springer papers	3 concepts	3 partial drafts	3 submissions or ready drafts	Draft PDFs
GitHub repositories	Org shell	2 repositories v0.1	3 repositories v1.0	GitHub links
LinkedIn/X posts	3 posts	6 additional posts	3–6 additional posts	Post links/analytics
Student teams	3 teams mapped	5 teams active	5–6 teams showcased	Team matrix
Collaboration meetings	Target list	2 meetings	3–5 meetings total	MoM documents
Demo videos	Scripts	1 draft video	3 published demos	Video links
Research newsletter	Template	Content draft	Issue 1 released	PDF/link
Research Day	Date fixed	Posters/demo plan	Event completed	Event report

17 Risk Register and Mitigation

Risk	Severity	Mitigation	Owner
Unrealistic publication timelines	High	Define success as submission/readiness, not acceptance; use weekly evidence gates.	Research Lead
Duplicate information sharing	High	Single tracker, single evidence folder, decision log, no status-only meetings.	Research Coordinator
Weak experimental evidence	High	Require baseline, ablation, validation and limitations before submission.	QA Board
Data privacy/ethics issues	High	Public-data statement or ethics clearance; no patient-identifiable data in public repositories.	Vertical Leads
Fragmented branding	Medium	One content calendar and approval workflow; branded templates only.	Branding Cell
Low faculty participation	Medium	Assign concrete paper roles; recognize contributors in monthly newsletter.	HoD/Research Lead
Student teams lose focus	Medium	2-week sprints, GitHub commits, demo milestones, mentor check-ins.	Faculty mentors

APC constraints	Medium	Maintain mixed venue strategy: Q1/Q2, IEEE/Springer conferences, no-APC journals where feasible.	Corresponding authors
Tool/platform delays	Low	Use temporary internal folders and publish once approvals are ready.	Branding Cell

18 Submission Quality Gate

✓ Scientific readiness

1. Clear problem statement and novelty.
2. Dataset source and split protocol documented.
3. Baselines are fair and recent.
4. Ablation study included.
5. Results are reproducible from archived scripts.
6. Limitations are honest and visible.

✓ Integrity readiness

1. Plagiarism/originality check completed.
2. AI-assisted writing carefully reviewed by authors.
3. Ethical/public data statement included.
4. No unsupported claims or invented results.
5. Author contributions finalized.
6. Conflicts of interest and funding statements added.

📄 Technical readiness

1. Tables match text claims.
2. Figures are high-resolution and numbered correctly.
3. Equations/algorithms are called out in text.
4. References are complete and current.
5. Journal template followed.
6. Supplementary files prepared.

🔄 Reproducibility readiness

1. README explains how to run code.
2. Environment file available.
3. Dataset access instructions included.
4. Model card or datasheet included.
5. License/citation file added.
6. Release tag created before submission.

19 Department Research Brochure Structure

Page	Section	Content
1	Cover	Department logo, DSU logo, motto, research identity and QR code to official department page.
2	Why CS&ME?	Short narrative connecting engineering, healthcare, AI-first DSU and translational impact.
3	Research verticals	Four verticals with icons, focus areas and flagship projects.

4	Publication pipeline	P1–P8 table, target outcomes and submission milestones.
5	Reproducible research	GitHub organization, repository standards, model cards and reproducibility checklist.
6	Student research ecosystem	Teams, mentorship model, hackathons, demos and project-to-paper pathway.
7	Collaboration opportunities	CDSIMER, AIISH/NIMHANS-style clinical engagement, MedTech partners, datasets, internships.
8	Contact and next steps	Research lead, HoD/Chairperson, LinkedIn/X/GitHub QR codes and call for collaboration.

20 August Research Day Blueprint

Event theme

CSME AI-for-Healthcare Research Day 2026: From Algorithms to Accessible Care

Segment	Duration	Output
Opening and vision	10 min	HoD/Research Lead frames CSME as AI-for-Healthcare hub.
Research vertical showcase	30 min	4 vertical leads present 5-minute roadmaps each.
Student demo round	45 min	5–6 teams demonstrate prototypes or posters.
Publication clinic	30 min	Review of paper pipeline and submission quality gates.
External expert talk	30 min	Clinical/industry perspective on translational AI.
GitHub and branding reveal	15 min	Repositories, newsletter and brochure introduced.
Closing and next actions	10 min	September–December milestone plan announced.

21 Final August Report Template

Report structure for HoD/Dean

1. Executive summary: 1 page.
2. Research Cell constitution and vertical-wise roles.
3. Publication pipeline: submitted, under preparation, planned.
4. Evidence dashboard: GitHub links, manuscript folders, submission receipts.
5. Branding dashboard: LinkedIn/X analytics, brochure, newsletter, demo videos.
6. Student research engagement: teams, mentors, outcomes.
7. Collaboration status: meetings, MoUs, next actions.
8. Financial/resource needs: APCs, devices, datasets, cloud/GPU needs.
9. Challenges and mitigation.
10. September–December 2026 roadmap.

22 Immediate Next 10 Actions

1. Approve the Research Cell Charter

Finalize membership, vertical leads, QA board and branding cell within 48 hours.

2. Open the Single Tracker

Create one official tracker and migrate all paper/project information into it.

3. Freeze P1–P8

Confirm titles, author teams, evidence folder and target venue class.

4. Create Evidence Folders

One folder per paper/project with manuscript, code, figures, tables, references and minutes.

5. Launch GitHub Organization

Create repository shells and repository standards before any code is uploaded.

6. Start Quality Gate Reviews

Assign two reviewers to each priority manuscript.

7. Launch LinkedIn/X

Use official bios, banners and first three posts.

8. Contact Collaborators

Schedule CDSIMER/clinical/industry meetings with formal minutes.

9. Schedule Research Day

Block late-August date, reserve venue, collect demo abstracts.

10. Prepare First Newsletter

Use CSME Research Pulse as the monthly visibility instrument.

23 Closing Statement

Strategic conclusion

By the end of August 2026, CS&ME should be able to demonstrate a credible research system, not only isolated research activity. The department's strongest 3-month outcome is a visible, disciplined, reproducible and branded research pipeline: two serious Q1/Q2 submissions, three IEEE/Springer/Scopus submissions or ready drafts, a working GitHub organization, active LinkedIn/X communication, student research teams, collaboration records, and one public-facing AI-for-Healthcare Research Day.